

① Calcular as integrais indefinidas:

$$\int \sqrt[4]{1-x} dx = \int u^{\frac{1}{4}} du = - \int u^{\frac{1}{4}} = - \frac{4}{5} u^{\frac{5}{4}} = - \frac{4(1-x)^{\frac{5}{4}}}{5} + C$$

$$u = 1-x$$

$$du = -1$$

$$\int x^{-\frac{2}{3}} dx = \frac{5x^{\frac{1}{3}}}{\frac{1}{3}} + C$$

$$\int \frac{2}{x^2} dx = 2 \int x^{-2} = 2 \frac{x^{-1}}{-1} = -\frac{2}{x} + C$$

② Calcular os seguintes integrais definidos:

$$\int_0^1 (x^2 + x) dx = \int_0^1 x^2 + \int_0^1 x = \left(\frac{1^3}{3} - \frac{0^3}{3} \right) + \left(\frac{1^2}{2} - \frac{0^2}{2} \right) \rightarrow$$

$$\rightarrow \frac{1}{3} + \frac{1}{2} = \frac{2}{6} + \frac{3}{6} = \frac{5}{6} \text{ u.a.}$$

$$\int_{\frac{\pi}{2}}^{\pi} \sin(x) dx = [-\cos(\pi)] - [-\cos(\frac{\pi}{2})] = [1] - [0] = 1 \text{ u.a.}$$

③ Achar a região limitada pelo parábola $y = -x^2 + 2$ e pelo eixo.

$$\int_{-\sqrt{2}}^{\sqrt{2}} (-x^2 + 2) dx = \int_{-\sqrt{2}}^{\sqrt{2}} -x^2 + \int_{-\sqrt{2}}^{\sqrt{2}} 2 = - \int_{-\sqrt{2}}^{\sqrt{2}} x^2 + \int_{-\sqrt{2}}^{\sqrt{2}} 2 \rightarrow - \frac{(x^3)}{3} + 2x \Big|_{-\sqrt{2}}^{\sqrt{2}}$$

$$\text{Upper limit} = \sqrt{2}$$

$$\text{Lower limit} = -\sqrt{2} \left[-\frac{2\sqrt{2}}{3} + 2\sqrt{2} \right] - \left[\frac{2\sqrt{2}}{3} - 2\sqrt{2} \right] = -\frac{4\sqrt{2}}{3} + 4\sqrt{2}$$

05/04/20

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④ Acha a área da região limitada por $y = 5 - x^2$ e $y = x + 3$.
Dica: achar os pontos de interseção e fazer gráfico.

$$\text{Área} = \int (5 - x^2) dx - \int (x + 3) dx = -\frac{x^3}{3} + 5x - \frac{x^2}{2} - 3x \rightarrow$$

$$\rightarrow -\frac{x^3}{3} + 5x - \frac{x^2}{2} - 3x \rightarrow -\frac{x^3}{3} + \frac{x^2}{2} + 8x + 2c$$

Upper limit: 1

Lower limit: -2

$\rightarrow F(x)$

Calculando ponto de interseção:

$$\rightarrow \frac{-9 \pm \sqrt{81}}{2} = -2$$

$$5 - x^2 = x + 3 \quad \Delta = b^2 - 4ac \quad x = \frac{-b \pm \sqrt{\Delta}}{2} = \frac{-1 \pm 3}{2}$$

$$5 - 3 = x + x^2 \quad \Delta = 1 \pm 8 \quad x = \frac{-1 \pm 3}{2} \rightarrow \frac{2}{2} = 1$$

$$x^2 + x - 2 = 0 \quad \Delta = 9 \quad y = x + 3$$

$$y = -2 + 3 = 1 \quad y = 1 + 3 = 4$$

Pontos = (-2; 1), (1; 4)

$$\int (5 - x^2) - \int (x + 3) \rightarrow$$

$$-2 \rightarrow \left[\frac{1^2}{2} + 3 \right] - \left[\frac{(-2)^2}{2} - 6 \right] = \left[\frac{1}{2} + 3 \right] - \left[2 - 6 \right] = \frac{7}{2} + 4 = \frac{15}{2}$$

$$\rightarrow \left[-\frac{1^3}{3} + 5 \right] - \left[-\frac{(-2)^3}{3} - 10 \right] = \left[-\frac{1}{3} + 5 \right] - \left[\frac{8}{3} - 10 \right] = \frac{14}{3} + \frac{22}{3} = \frac{36}{3} = 12$$

$$\rightarrow \frac{-1}{3} + \frac{15}{3} - \frac{8}{3} + \frac{30}{3} = \frac{36}{3} = 12$$

$$\text{Resultado} = 12 - \frac{15}{2} = \frac{24}{2} - \frac{15}{2} = \frac{9}{2} \text{ u.a}$$

